

Principles of Microeconomics — Day 3

In-Class Problem Set: Monopoly

This problem set accompanies the Day 3 session. Numbers differ from the lecture example. Solutions (instructor copy) are collected at the end; set `\solnfalse` in the preamble to produce a student version.

Exercise 1 — Monopoly, deadweight loss and price discrimination

A monopolist faces demand $P = 100 - 2Q$ and has constant marginal cost $MC = 20$ (no fixed cost).

- (a) Find the monopoly quantity Q^m , price P^m and profit.
- (b) Find consumer surplus and the deadweight loss relative to the competitive benchmark.
- (c) Under *perfect* (first-degree) price discrimination, find the output, total surplus, consumer surplus and deadweight loss.
- (d) The firm can instead split buyers into two groups: market 1 with $P_1 = 100 - 2Q_1$ and market 2 with $P_2 = 80 - 2Q_2$, same $MC = 20$. Find each price and quantity, and state which group pays more and why.

Exercise 2 — A monopolist with rising marginal cost

A monopolist faces demand $P = 240 - 4Q$ and has total cost $TC(Q) = 2Q^2$ (so $MC = 4Q$, no fixed cost).

- (a) Find the monopoly quantity Q^m , price P^m and profit.
- (b) Find consumer surplus, the competitive-benchmark quantity, and the deadweight loss.
- (c) A fixed cost of 500 is now added to the firm's costs. Does the profit-maximising quantity change? Does profit change? Explain in one sentence why.